

(Unit No. 1: Matrices)

Q.1 Do as directed.

1. Do the system of linear equation $x + y + z = 0$, $3x + 3y - z = 0$ has a unique solution?
 → (b) No

2. If these linear equations in a system with two variables represent parallel lines then the system has a unique solution.

→ (a) False

3. The system of linear equations $ax - 6y + 2 = 0$, $ax - 3y = 0$ do not have unique solution of $a = 2$

$$\rightarrow \frac{a}{a} = \frac{+6}{+3} \quad \therefore a = \frac{+6 \times 3}{3}$$

$$\boxed{\therefore a = 2}$$

4. The solution of the system of linear equation $x = 0$, $2y = 4$ is the point
 → (c) (0, 2)

5. The matrix equation representing the system of linear equations, $2x + 4y - 3 = 0$, $5x + 2y + 2 = 0$, is

$$\rightarrow (d) \begin{bmatrix} 2 & 4 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

Q.2 DO as directed (3/4 marks)

1. Solve the system of linear equations $3x + y = 9$, $2x - 3y = -5$ using matrices.

→

The given system of linear eqⁿs can be written as

$$A \cdot X = B$$

Where

$$A = \begin{bmatrix} 3 & 1 \\ 2 & -3 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$B = \begin{bmatrix} 9 \\ -5 \end{bmatrix}$$

Now,

$$A = \begin{bmatrix} 3 & 1 \\ 2 & -3 \end{bmatrix}$$

$$\begin{aligned} |A| &= (3)(-3) - (2)(1) \\ &= -9 - 2 \\ &= -11 \neq 0 \end{aligned}$$

$\therefore A^{-1}$ exists

$$A^{-1} = \frac{1}{|A|} \text{Adj}(A)$$

Note,

$$A = \begin{bmatrix} 5 & 2 \\ 3 & 1 \end{bmatrix}$$

$$\text{Adj}(A) = \begin{bmatrix} -3 & -1 \\ -2 & 3 \end{bmatrix}$$

We have,

$$AX = B$$

$$X = A^{-1} \cdot B$$

$$= \frac{1}{|A|} \cdot \text{Adj}(A) \cdot B$$

$$= \frac{1}{11} \cdot \begin{bmatrix} -3 & -1 \\ -2 & 3 \end{bmatrix} \cdot \begin{bmatrix} 9 \\ -5 \end{bmatrix}$$

$$= \frac{1}{11} \cdot \begin{bmatrix} (-3)(9) + (-1)(-5) \\ (-2)(9) + (3)(-5) \end{bmatrix}$$

$$= \frac{1}{11} \cdot \begin{bmatrix} -27 + 5 \\ -18 - 15 \end{bmatrix}$$

$$= \frac{1}{11} \cdot \begin{bmatrix} -22 \\ -33 \end{bmatrix}$$

$$= \begin{bmatrix} -22/11 \\ -33/11 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

2. Solve the system of linear equations $2x + 3y = 1$, $y - 4x = 2$ using matrices.

→

The given system of linear eqⁿs can be written as:

$$A \cdot X = B$$

Where,

$$A = \begin{bmatrix} 2 & 3 \\ -4 & 1 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$B = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Now,

$$A = \begin{bmatrix} 2 & 3 \\ -4 & 1 \end{bmatrix}$$

$$|A| = (2)(1) - (-4)(3) = 2 + 12 = 14 \neq 0$$

Note, $A = \begin{bmatrix} 2 & 3 \\ -4 & 1 \end{bmatrix}$

$$\text{Adj}(A) = \begin{bmatrix} 1 & -3 \\ 4 & 2 \end{bmatrix}$$

We have,

$$A \cdot X = B$$

$$X = A^{-1} \cdot B$$

$$= \frac{1}{|A|} \cdot \text{Adj}(A) \cdot B$$

$$= \frac{1}{14} \cdot \begin{bmatrix} 1 & -3 \\ 4 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$= \frac{1}{14} \begin{bmatrix} (1)(1) + (-3)(2) \\ (4)(1) + (2)(2) \end{bmatrix}$$

$$= \frac{1}{14} \begin{bmatrix} 1 - 6 \\ 4 + 4 \end{bmatrix}$$

$$= \frac{1}{14} \begin{bmatrix} -5 \\ 8 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -5/14 \\ 8/14 \end{bmatrix} = \begin{bmatrix} -5/14 \\ 4/7 \end{bmatrix}$$

3. Solve the system of linear equations $\frac{3}{x} + \frac{2}{y} + z = 0$
 $\frac{1}{x} - \frac{4}{y} + 6 = 0$ using matrices.

→ The given system of linear eqⁿs can be written as $A \cdot X = B$

where,

$$A = \begin{bmatrix} 3 & 2 \\ 1 & -4 \end{bmatrix}$$

$$X = \begin{bmatrix} \frac{1}{3}x \\ \frac{1}{2}y \end{bmatrix}$$

$$B = \begin{bmatrix} -2 \\ -6 \end{bmatrix}$$

Now, $A = \begin{bmatrix} 3 & 2 \\ 1 & -4 \end{bmatrix}$

$$|A| = (3)(-4) - (1)(2) = -12 - 2 = -14 \neq 0$$

Note, $A = \begin{bmatrix} 3 & 2 \\ 1 & -4 \end{bmatrix}$

$$\text{Adj}(A) = \begin{bmatrix} -4 & -2 \\ -1 & 3 \end{bmatrix}$$

We have,

$$A \cdot X = B$$

$$X = A^{-1} \cdot B$$

$$= \frac{1}{|A|} \cdot \text{Adj}(A) \cdot B$$

$$= \frac{1}{-14} \cdot \begin{bmatrix} -4 & -2 \\ -1 & 3 \end{bmatrix} \cdot \begin{bmatrix} -2 \\ -6 \end{bmatrix}$$

$$= \frac{1}{-14} \cdot \begin{bmatrix} (-4)(-2) + (-2)(-6) \\ (-1)(-2) + (3)(-6) \end{bmatrix}$$

$$= \frac{1}{-14} \begin{bmatrix} 8 + 12 \\ 2 - 18 \end{bmatrix}$$

$$= \frac{1}{-14} \cdot \begin{bmatrix} 20 \\ 16 \end{bmatrix}$$

$$= \begin{bmatrix} -20/14 \\ -16/14 \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{x} \\ \frac{1}{y} \end{bmatrix} = \begin{bmatrix} -10/7 \\ -8/7 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -7/10 \\ -7/8 \end{bmatrix}$$

$$\therefore x = -7/10, y = -7/8$$

4. Solve the system of linear eqⁿs $2x + 3y = 6xy$ & $x - y = xy$ using matrices.

→ Here,

$$\begin{aligned} 2x + 3y &= 6xy \\ x - y &= xy \end{aligned}$$

दोनों eqⁿs मा = दोनों बाजूया x y से लाया।

$$\therefore \frac{2x}{xy} + \frac{3y}{xy} = \frac{6xy}{xy}$$

$$\therefore \frac{2}{y} + \frac{3}{x} = 6$$

$$\therefore 3\left(\frac{1}{x}\right) + 2\left(\frac{1}{y}\right) = 6 \quad \text{--- (1)}$$

$$\frac{x}{xy} - \frac{y}{xy} = \frac{xy}{xy}$$

$$\therefore \frac{1}{y} - \frac{1}{x} = 1$$

$$\therefore -\frac{1}{x} + \frac{1}{y} = 1 \quad (2)$$

The Collected System of linear eq^{ns} can be written as

$$A \cdot X = B$$

Where,

$$A = \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix}$$

$$X = \begin{bmatrix} \frac{1}{x} \\ \frac{1}{y} \end{bmatrix}$$

$$B = \begin{bmatrix} 6 \\ 1 \end{bmatrix}$$

Now, $A = \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix}$

$$|A| = (3)(1) - (-1)(2) = 3 + 2 = 5 \neq 0$$

Note, $A = \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix}$

$$\text{Adj}(A) = \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}$$

We have,

$$A \cdot X = B$$

$$X = A^{-1} \cdot B$$

$$= \frac{1}{|A|} \cdot \text{Adj}(A) \cdot B$$

$$= \frac{1}{5} \cdot \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix} \cdot \begin{bmatrix} 6 \\ 1 \end{bmatrix}$$

$$= \frac{1}{5} \cdot \begin{bmatrix} (1)(6) + (-2)(1) \\ (1)(6) + (3)(1) \end{bmatrix}$$

$$= \frac{1}{5} \cdot \begin{bmatrix} 6 - 2 \\ 6 + 3 \end{bmatrix}$$

$$= \frac{1}{5} \begin{bmatrix} 4 \\ 9 \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{5}x \\ \frac{1}{5}y \end{bmatrix} = \begin{bmatrix} \frac{4}{5} \\ \frac{9}{5} \end{bmatrix}$$

$$\therefore \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{5}{5} \cdot \frac{4}{5} \\ \frac{5}{5} \cdot \frac{9}{5} \end{bmatrix}$$

$$\therefore x = \frac{5}{5} \cdot \frac{4}{5} ; y = \frac{5}{5} \cdot \frac{9}{5}$$